



Laboratory work №2

Databases

Tolepказы Galamat

Tuesday, 11:00

Almaty 2026

Task 1. Identify the normal form violations

```
Booking_receipt(  
  booking_id,  
  passenger_full_name,  
  passenger_passport_number,  
  flight_number,  
  departure_airport_name,  
  departure_city,  
  arrival_airport_name,  
  arrival_city,  
  airline_name,  
  seat_numbers,  
  ticket_price  
)
```

The table violates the First Normal Form (1NF) because the seat_numbers column can contain multiple values in a single cell. For example, one cell can contain "12A, 12B". In 1NF, each attribute must contain only one atomic value. Therefore, seat_numbers must be separated into individual values.

Task 2. Step 1 — First Normal Form (1NF)

booking_id	passenger_full_name	passenger_passport_number	flight_number	departure_airport_name	departure_city	arrival_airport_name	arrival_city	airline_name	seat_number	ticket_price
1001	John Smith	P001	KC101	Almaty Airport	Almaty	Astana Airport	Astana	Air Astana	12A	50000
1001	Jane Smith	P002	KC101	Almaty Airport	Almaty	Astana Airport	Astana	Air Astana	12B	50000
1002	Alex Brown	P003	FS202	Astana Airport	Astana	Almaty Airport	Almaty	FlyArystan	15C	35000

Primary Key: (booking_id, passenger_passport_number)

Step 2 — Second Normal Form (2NF)

The 1NF table has a composite primary key consisting of booking_id and passenger_passport_number. However, some non-key attributes depend only on part of the composite key. For example, booking_id determines flight_number and ticket_price, while passenger_passport_number determines passenger_full_name. These are partial dependencies, so the table is not in 2NF.

<pre>Booking(<u>booking_id</u>, flight_number, ticket_price)</pre>	<pre>Passenger(<u>passenger_passport_number</u>, passenger_full_name)</pre>	<pre>Booking_Passenger(<u>booking_id</u>, <u>passenger_passport_number</u>, seat_number)</pre>
--	---	--

Result: The partial dependencies have been removed by separating booking information and passenger information into different tables. The Booking_Passenger table stores the relationship between bookings and passengers. Therefore, the resulting tables are in 2NF.

Step 3 — Third Normal Form (3NF)

Flight(
 flight_number,
 departure_airport_id,
 arrival_airport_id,
 airline_id
)

flight_number
|
| $\vdash$ *departure_airport_id*
| ↓
| *Airport*
| $\vdash$ *airport_name*
| └ *city*
|
| $\vdash$ *arrival_airport_id*
| ↓
| *Airport*
| $\vdash$ *airport_name*
| └ *city*
|
└ *airline_id*
 ↓
 Airline
 └ *airline_name*

Final 3NF Structure

Airline
└— PK *airline_id*
└ *airline_name*

Airport
└— PK *airport_id*
└ *airport_name*
└ *city*

Flight
└— PK *flight_number*
└— FK *departure_airport_id*
└— FK *arrival_airport_id*
└ FK *airline_id*

Booking
└— PK *booking_id*
└— FK *flight_number*
└ *ticket_price*

Passenger

└─ PK passenger_passport_number
└─ passenger_full_name

Booking_Passenger

└─ PK booking_id
└─ PK passenger_passport_number
└─ FK booking_id
└─ FK passenger_passport_number
└─ seat_number

Result: The transitive dependencies were removed by separating flight, airline and airport information into different tables. Each non-key attribute now depends on the key of its own table and not on another non-key attribute. Therefore, the resulting database structure satisfies the Third Normal Form (3NF).

Task 3 — Functional Dependencies

1NF

$(booking_id, passenger_passport_number) \rightarrow seat_number$

2NF

$booking_id \rightarrow flight_number$

$booking_id \rightarrow ticket_price$

$passenger_passport_number \rightarrow passenger_full_name$

3NF

$flight_number \rightarrow airline_id$

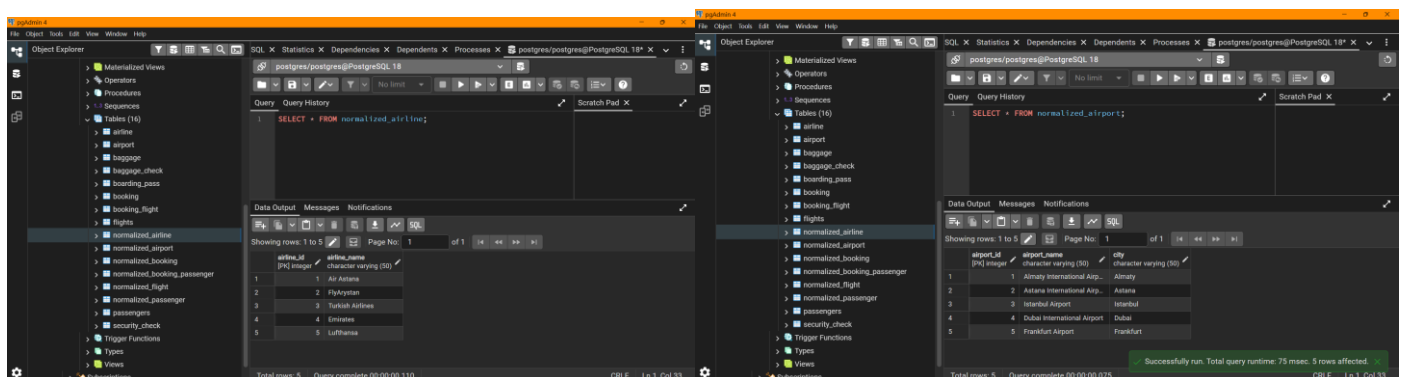
$flight_number \rightarrow departure_airport_id$

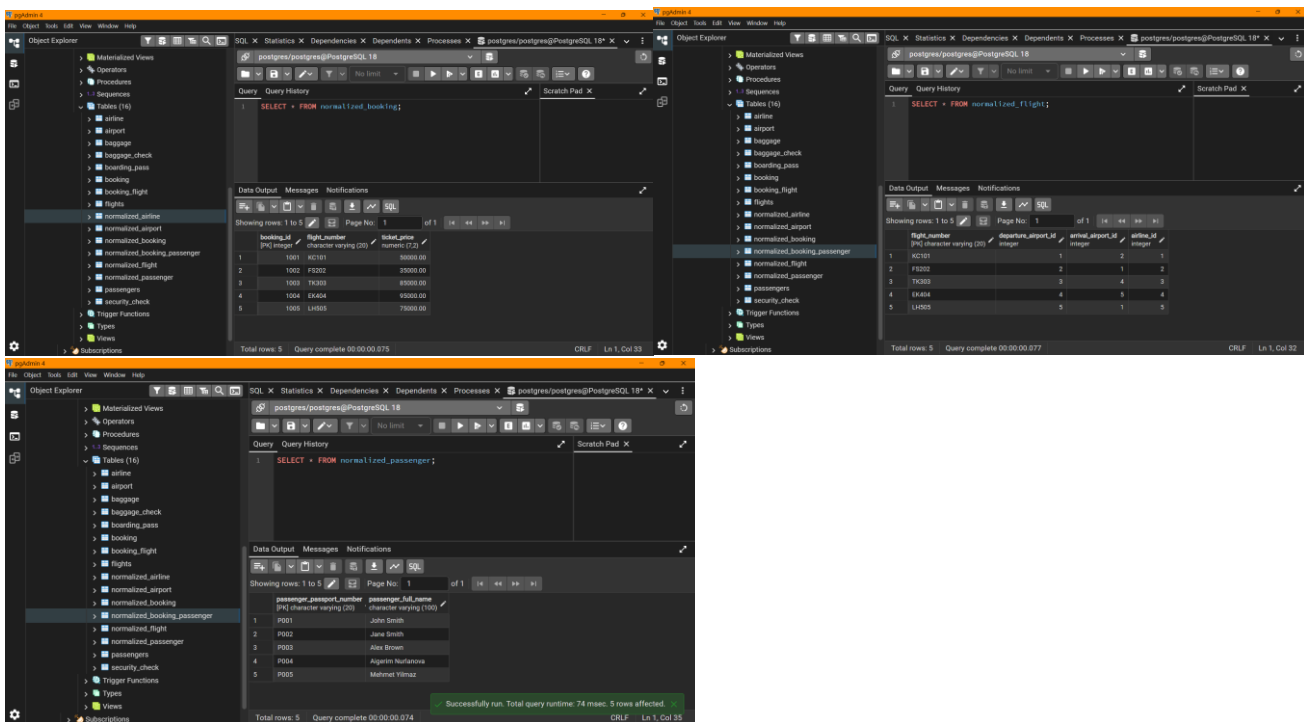
$flight_number \rightarrow arrival_airport_id$

$airline_id \rightarrow airline_name$

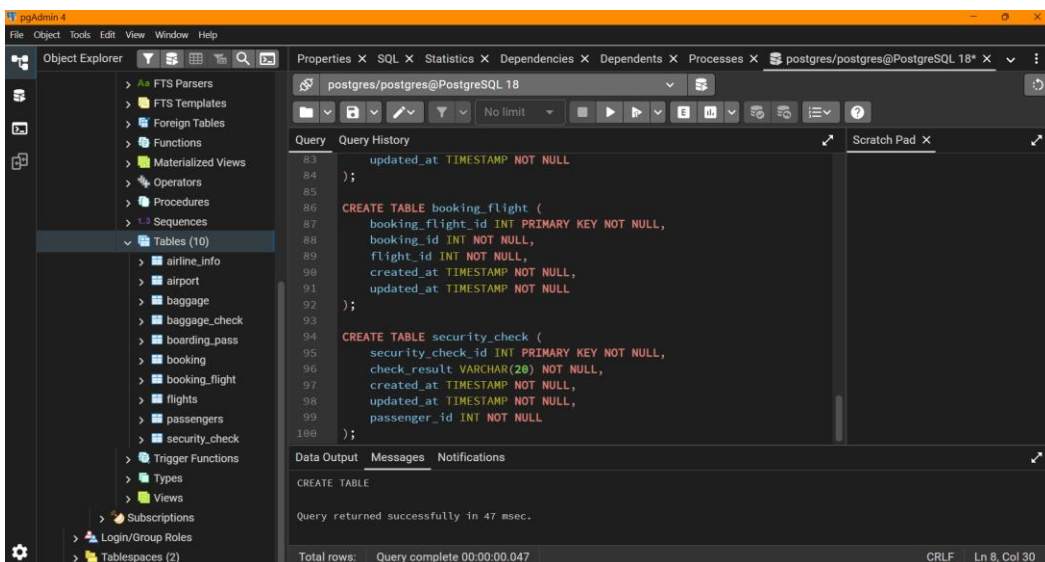
$airport_id \rightarrow airport_name$

$airport_id \rightarrow city$

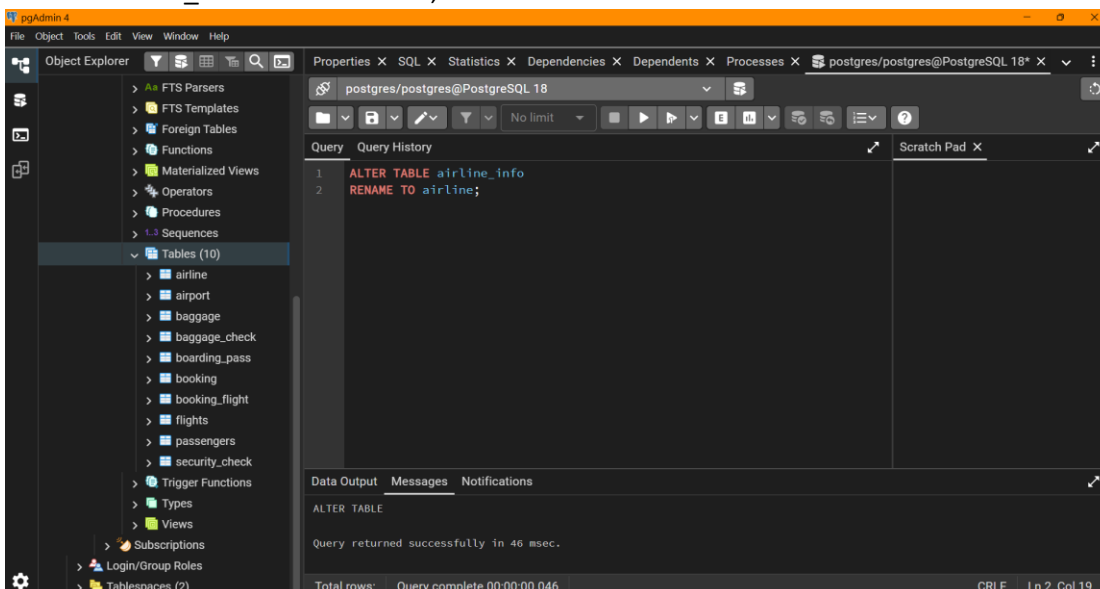




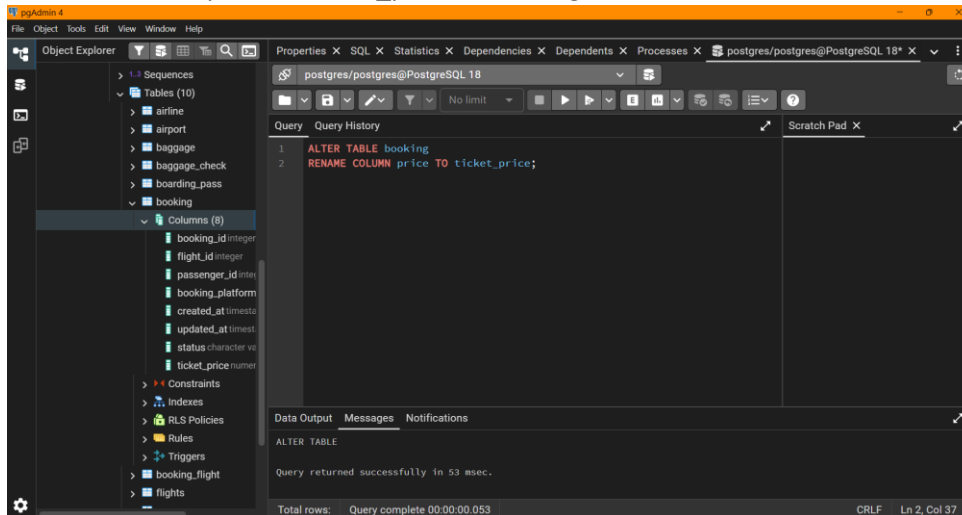
2. Define Primary Keys for each tables;
4. Define for all attributes not null constraint;



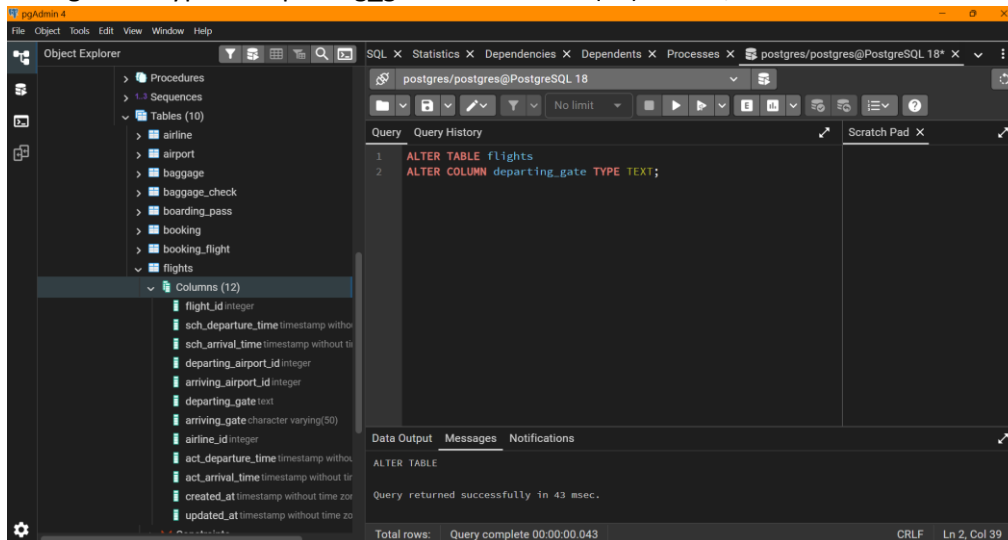
Rename airline_info table to airline;



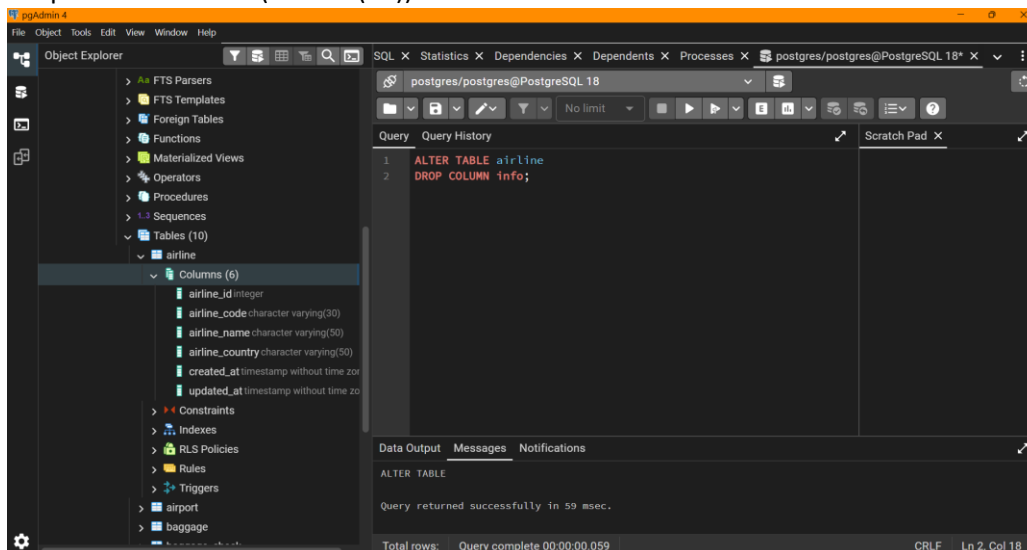
Rename column price to ticket_price in booking table;



Change data type of departing_gate from varchar(50) to text;



Drop the column info(varchar(50)) from the airline table.



Make a relationship between following tables:

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File Object Tools Edit View Window Help

Object Explorer

- airline
 - Columns
 - Constraints (1)
 - airline_info_pkey
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
- airport
 - Columns (7)
 - Constraints (1)
 - airport_pkey
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
- baggage
 - Columns (5)
 - Constraints (2)
 - baggage_pkey
 - fk_baggage_booking
 - Indexes
 - RLS Policies
 - Rules
 - Triggers

SQL X Statistics X Dependencies X Dependents X Processes X postgres/postgres@PostgreSQL 18* X

postgres/postgres@PostgreSQL 18

Query Query History Scratch Pad X

```
1 ALTER TABLE flights
2 ADD CONSTRAINT fk_flights_airline
3 FOREIGN KEY (airline_id)
4 REFERENCES airline(airline_id);
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 43 msec.

Total rows: Query complete 00:00:00.043 CRLF Ln 4, Col 32

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File Object Tools Edit View Window Help

Object Explorer

- baggage_check
 - Columns
 - Constraints (3)
 - baggage_check_pkey
 - fk_baggage_check_booking
 - fk_baggage_check_passenger
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
- boarding_pass
 - Columns
 - Constraints (2)
 - boarding_pass_pkey
 - fk_boarding_pass_booking
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
- booking
 - Columns
 - Constraints (2)
 - booking_pkey
 - fk_booking_passenger
 - Indexes

SQL X Statistics X Dependencies X Dependents X Processes X postgres/postgres@PostgreSQL 18* X

postgres/postgres@PostgreSQL 18

Query Query History Scratch Pad X

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```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 43 msec.

Total rows: Query complete 00:00:00.043 CRLF Ln 4, Col 32

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- booking_flight
 - Columns
 - Constraints (3)
 - booking_flight_pkey
 - fk_booking_flight_booking
 - fk_booking_flight_flight
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
- flights
 - Columns (12)
 - Constraints (4)
 - fk_flights_airline
 - fk_flights_arriving_airport
 - fk_flights_departing_airport
 - flights_pkey
 - Indexes
 - RLS Policies
 - Rules
 - Triggers
- passengers
 - Columns (10)
 - Constraints (1)
 - passengers_pkey
 - Indexes

SQL X Statistics X Dependencies X Dependents X Processes X postgres/postgres@PostgreSQL 18* X

postgres/postgres@PostgreSQL 18

Query Query History Scratch Pad X

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2 ADD CONSTRAINT fk_flights_airline
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Data Output Messages Notifications

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